Haoran (Jensen) Yuan

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EDUCATION

Doctor of Philosophy in Computer Engineering — Aug. 2026 – May 2028 (Expected)
University of Illinois, Urbana-Champaign — Champaign, IL

Master of Science in Computer Engineering — Sept. 2024-May. 2026
University of Illinois, Urbana-Champaign — Champaign, IL

Bachelor of Science in Electrical Engineering, Minor in Computer Science — Sept. 2020 – Jun. 2024
University of Illinois, Urbana-Champaign |GPA:3.83/4.0 — Champaign, IL

Bachelor of Engineering in Computer Engineering — Sept. 2020 – Jun. 2024
Zhejiang University | GRA 3.96/4.0 — Hangzhou, China

RESEARCH EXPERIENCE

DexTacWAM: Visuo-Tactile World-Action Model (Project Lead, BAIR @ UC Berkeley) | *World Model, Video DiT, Tactile*

- Led a visuo-tactile **World-Action Model** for dexterous manipulation on a 22-DoF bimanual platform with 10 fingertip tactile sensors, modeling hidden contact dynamics (pressure, slip) that cameras miss under occlusion; featured by BAIR.
- Designed a finger- and pose-aware **tactile compressor** (self-attention pooling + TimeSformer refinement) fusing 10 fingertip latents into 2 hand views, retaining **89.4%** of contact recall (vs. 62.9% for cross-attention pooling) and cutting DiT tokens by **3.67×**.
- Extended a pretrained video DiT with tactile views via a **modality-split flow-matching** objective and an action expert with per-modality K/V RMS normalization; 3-stage pipeline needs only ∼100 demos per task, with **2.26×** faster training and **1.29×** faster inference.
- Achieved **70.6%** avg. success on 6 real-robot tasks vs. **38.0%** for the best baseline (RDP); **74.7%** vs. 26.6% w/o tactile world modeling.

JetSpec: Parallel Tree Drafting for Speculative Decoding (Core Contributor, Hao AI Lab @ UCSD) | *Speculative Decoding, Triton, vLLM*

- Co-developed a **causal parallel draft head** that emits a complete scored speculative tree in **one forward pass** while preserving the target model's autoregressive factorization along every branch.
- Standalone inference engine with paged FlashAttention **tree-attention kernels** in Triton and CuTe DSL (no dense mask materialization), plus vLLM integration.
- Achieved up to **9.64×** end-to-end speedup over AR on Qwen3-8B (MATH-500, H100), beating DDTree (8.78×) and EAGLE-3; sustains ∼1,000 tok/s on a single B200.

VTAM: Video–Tactile Action Model (Project Lead, PLAN Lab @ UIUC) | *PyTorch, World Model, Multimodal Learning*

- Led end-to-end development (2k+ real-world trajectories, model design, training, deployment), achieving **90% success** on average on chip pick-and-place, plate wiping and cucumber peeling.
- Designed a gated cross-attention mechanism to suppress cross-modal contamination between tactile and visual tokens by sparsifying attention, reducing training loss spikes by **42%** and improving convergence speed by **1.7×**.
- Introduced a force-aware regularized diffusion objective to balance tactile–vision attention, raising success by **+80% over vision-only**.
- Built a parallelized three-stage training pipeline, cutting training time from **∼50h to ∼30h on 4×A100** and token usage by **25%**.

JetSpec++: Foresight Parallel Drafting with Self Refinements (Core Contributor, Hao AI Lab @ UCSD) | *Speculative Decoding*

- Developed JetSpec++, a **causal correction for parallel drafting** that conditions each drafted token on its **full branch history** rather than on the preceding token alone, with foresight drafting that pre-drafts the next block during verification.
- At matched parameter count, **outperformed DeepSeek's DSpark**, with the margin widening at draft depths beyond those used in training; raised average acceptance length to **8.94** (vs. 6.89) across 7 benchmarks.
- Achieved up to **7.45×** speedup over AR and **1.54×** over DSpark on Qwen3-8B, reaching **1,361 tok/s** for a single request on GB200.

WORK EXPERIENCE

Snowflake: | Software Engineering Intern — Menlo Park, California | **May 2025 –August 2025**

- Designed and implemented a **predictive autoscaler** for Snowflake GS clusters using a combination of Gradient Boosting Machine, ARIMA and XGBoost models trained on historical telemetry via Snowhouse database.
- Built a SQL-based forecasting engine for on-the-fly CPU workload prediction without pretraining or model maintenance overhead.
- Supported **multi-cluster, multi-deployment** environments; executed end-to-end model training and predictions in <5 minutes per deployment.
- Achieved **14.3% MAPE** and **92.5% coverage** at the **95% confidence interval** for **24-hour** CPU utilization forecasts in **one execution** using less than one month of historical data and verified on **all active deployments**.
- Deployed on the internal background job system to dynamically adjust instance provisioning, increasing average **CPU utilization by >20%** and reducing overprovisioning in simulated datacenter environments.

TECHNICAL SKILLS

Languages: Python, C/C++, Rust, Java, SQL, Systemverilog.
Frameworks: SGLang, PyTorch, CUDA, Isaac Gym, Isaac Sim, vLLM, Mujoco, Flask, SQLAlchemy, OpenCV.
Database & Tools: MySQL, PostgreSQL, Redis, Docker, AWS, GCP, Git, Linux, Shell.